



Top 5 PreTrained Models in Natural Language Processing (NLP)

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Pretrained models are deep learning models that have been trained on huge amounts of data before fine-tuning for a specific task. The pre-trained models have revolutionized the landscape of **natural language processing** as they allow the developer to transfer the learned knowledge to specific tasks, even if the tasks differ from the original training data. The article aims to provide a comprehensive knowledge of pre-trained models that are the driving force behind smart **NLP-based** the **AI** models like ChatGPT, Gemini, Bard and more.

What is a pre-trained model?

A pre-trained model, having been trained on extensive data, serves as a foundational model for various tasks, leveraging its learned patterns and features. In [natural language processing \(NLP\)](#), these models are commonly employed as a starting point for tasks like language translation, sentiment analysis, and text summarization. Utilizing pre-trained models allows NLP practitioners to economize on time and resources, bypassing the need to train a model from scratch on a large dataset.

Some popular pre-trained models for NLP include BERT, GPT-2, ELMo, and RoBERTa. These models are trained on large datasets of text and can be fine-tuned for specific tasks.

Why do we use Pretrained Models?

Pretrained models are effective and efficient solutions for developers, researchers and businesses as they eliminate the need to write and train the code from scratch, saving time. Other advantages of using pre-trained models for projects are:

- Reduces the computational burden required for initial model training hence, making development more accessible.
- The learned knowledge can be used for various applications.
- Models can be fine-tuned according to the task and can result in superior performance to training from the initial point.
- Less labelled data is required for fine-tuning specific tasks.

Applications of pre-trained Models in NLP

- [Language translation](#): NLP algorithms facilitate the automatic translation of text or speech, promoting communication across language barriers.
- [Sentiment analysis](#): NLP algorithms enable automated analysis of text or speech to discern sentiment (positive, negative, or neutral), beneficial for tasks like customer feedback assessment and social media surveillance.

- Chatbot development: NLP algorithms can be used to develop chatbots that can understand and respond to human language. This is useful for applications such as customer service, where chatbots can handle routine inquiries and free up human agents to handle more complex tasks.
- [Text summarization](#): NLP algorithms can be used to automatically summarize long documents or articles by extracting the most important information and presenting it in a condensed form. This is useful for applications such as news aggregation and information extraction.
- Sentence completion: NLP algorithms can automatically generate sentence or paragraph continuations by considering the context and content of the entered text.

The application of pretrained models is not limited to NLP, it is also used for [image classification](#), [image segmentation](#) and other [computer vision](#) applications.

Pretrained models in NLP

Here are a few excellent pretrained models for natural language processing (NLP):

1. BERT (Bidirectional Encoder Representations from Transformers)

BERT (Bidirectional Encoder Representations from Transformers) is a state-of-the-art language representation model developed by Google. It is trained on a large dataset of unannotated text and can be fine-tuned for a wide range of natural language processing (NLP) tasks. BERT has achieved state-of-the-art performance on a variety of NLP tasks, such as language translation, sentiment analysis, and text summarization.

Architecture and Working

- The processes the textual sequence bidirectionally, considering both left and right context simultaneously for each word. This allows BERT to better capture the meaning and context of words in a sentence, leading to improved performance on a variety of NLP tasks.
- BERT is a transformer-based model, which means it uses self-attention mechanisms to process input text.
- The model is trained on extensive amount of dataset with diverse range of text. The pretraining phase allows the model to learn and understand the linguistic patterns.
- The BERT uses attention mechanism to assign different weights to different parts of input sequence.

To know more about BERT, you can refer to following links:

- [Explanation of BERT Model – NLP](#)
- [Understanding BERT – NLP](#)
- [Sentiment Classification Using BERT](#)
- [How to Generate Word Embedding using BERT?](#)
- [Fine-tuning BERT model for Sentiment Analysis](#)

2. GPT-2 (Generative Pretrained Transformer 2)

GPT-2 is a transformer-based model pretrained on an extensive English corpus in a self-supervised manner. This language model is developed by OpenAI. It is trained on a massive dataset of

unannotated text and can generate human-like text and perform various natural language processing (NLP) tasks. The smallest version of GPT-2 has 124 million parameters.

In detail, input sequences consist of continuous text of a defined length, with the corresponding targets being the same sequence shifted by one token. To ensure accurate predictions, the model internally employs a mask mechanism, restricting its focus to inputs only up to the current token and excluding future tokens. This sophisticated training process enables the model to learn an intrinsic representation of the English language, yielding features that prove valuable for downstream tasks.

In addition to text generation, GPT-2 can also be fine-tuned sentiment analysis and text classification problems.

3. ELMo (Embeddings from Language Models)

ELMo (Embeddings from Language Models) is a deep contextualized word representation model developed by researchers at the Allen Institute for Artificial Intelligence. It is trained on a large dataset of unannotated text and can be fine-tuned for a wide range of natural language processing (NLP) tasks. ELMo word vectors are generated through a two-layer bidirectional language model (biLM), featuring both forward and backward passes in each layer. Diverging from approaches like Glove and Word2Vec, ELMo takes a holistic perspective by representing word embeddings based on the entire sentence that encompasses the word. This unique characteristic enables ELMo embeddings to effectively capture the contextual nuances of a word within a given sentence. Consequently, ELMo has the capability to produce distinct embeddings for the same word deployed in diverse contexts across different sentences, setting it apart in its ability to grasp the intricacies of language use.

4. Transformer-XL

Transformer-XL is a state-of-the-art language representation model developed by researchers at Carnegie Mellon University and Google Brain. Transformer -XL is a variant of transformer model, which includes relative positional encoding and a recurrence mechanism. Transformers XL tackles the challenge of long-term dependency by retaining the previously learned segment in a hidden state. This means that instead of recalculating each segment's hidden state from scratch, the model utilizes the existing knowledge from the preceding segment for the current one. This innovative approach not only mitigates issues inherent in the vanilla transformer model but also effectively addresses the long-term dependency problem.

One notable advantage of Transformers XL lies in its ability to overcome context fragmentation. By avoiding the use of recently initialized or empty context information, the model ensures a more coherent understanding of context. This breakthrough allows the model to be applied seamlessly to character-level language modeling as well as word-level modeling, showcasing its versatility and enhanced performance in capturing intricate linguistic patterns.

Transformer-XL can be fine-tuned for a wide range of NLP tasks, including language translation, sentiment analysis, and text summarization.

5. RoBERTa (Robustly Optimized BERT)

RoBERTa (Robustly Optimized BERT) is a variant of BERT (Bidirectional Encoder Representations from Transformers) developed by researchers at Facebook AI. It is trained on a larger dataset and fine-tuned on a variety of natural language processing (NLP) tasks, making it a more powerful language representation model than BERT. RoBERTa is a transformer-based model, which means it uses self-attention mechanisms to process input text.

Like BERT, RoBERTa is "bidirectional," meaning it considers the context from both the left and the right sides of a token, rather than just the left side as in previous models. This allows RoBERTa to better capture the meaning and context of words in a sentence, leading to improved performance on a variety of NLP tasks. It has achieved state-of-the-art performance on several benchmarks, making it a powerful tool for NLP practitioners.

To know more about RoBERTa Model, check:

- [Overview of ROBERTa model](#)

Conclusion

In conclusion, pretrained models in NLP, such as BERT, GPT-2, ELMo, Transformer-XL, and RoBERTa, have revolutionized language understanding and application development. These models, trained on extensive datasets, provide a foundational basis for various NLP tasks, offering efficiency and superior performance.



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